Fold-compatible Ulam quotients and the deterministic spike barrier

RH research program

August 2026

Abstract

We audit the first continuation left open by RH-367 for the postcritically finite quadratic map $f(x) = 1-ux^2$, where $u^3-2u^2+2u-2 = 0$ and $J = [-(u-1), 1]$. Folding by $q(x) = |x|$ gives $T(y) = |1-uy^2|$ on $[0, 1]$. For partitions that are genuinely compatible with this fold, the exact full cell-overlap matrix has a folded quotient and an annihilated mirror kernel; its characteristic polynomial is z^m times the folded characteristic polynomial. Conditional expectations also give a real L^1 approximation and exterior resolvent convergence on $|z| > 1$. These facts do not produce a Riesz contour around -1. Indeed, on the natural bounded-variation strong space the deterministic projection of the terminal square-root spike has an adjacent-cell jump $(2-\sqrt{2})/\sqrt{uh}$, so its strong norm grows as $h^{-1/2}$. The positive strong-space bridge therefore remains a scoped negative. The exact finite quotient, the exterior weak bridge, and the spike obstruction form a standalone theorem edge. No continuum spectral limit, prime trace, Hilbert–Polya operator, or proof of RH is claimed.

1 Scope and frozen source

The source package fixes the real root

$$u = 1.543689012692076\ldots, \qquad u^3-2u^2+2u-2 = 0, \qquad r = u-1, \tag{1}$$

and the invariant interval and map

$$J = [-r, 1], \qquad f(x) = 1-ux^2. \tag{2}$$

The postcritical relations are $f(0) = 1$, $f(1) = -r$, $f(-r) = r$, and $f(r) = r$. The RH-367 source proves the two-band exchange

$$\mathcal{B}_0 = [-r, r], \qquad \mathcal{B}_1 = [r, 1], \qquad f(\mathcal{B}_0) = \mathcal{B}_1, \quad f(\mathcal{B}_1) = \mathcal{B}_0. \tag{3}$$

Put $q(x) = |x|$. The factor map and its folded representative are

$$q \circ f = T \circ q, \qquad T(y) = |1-uy^2|, \qquad 0 \le y \le 1. \tag{4}$$

The source locks used by the executable artifact are the cyclic-Ulam source and RH-367 [3, 1], together with RH-14, RH-52, RH-55, and the frozen four-volume verification [4, 2, 5]. The hashes and commits are recorded in `results/result.json`. In particular, the finite matrices below are not reconstructed from a numerical eigensolver.

Remark 1.1 (Data-type firewall)**.** *RH-367 proves a finite aligned sign mode. It does not identify that mode with an isolated continuum eigenvalue. RH-14 and RH-52 work on a declared tower/spike space, while RH-52 and RH-55 use a positive-noise schedule. This paper keeps the deterministic endpoint, the folded quotient, and the noisy results separate.*

2 Exact fold-compatible finite quotients

Definition 2.1 (Exact cell-overlap matrices). *Let $\mathcal{C} = \{C_i\}$ be an interval partition of $[0,1]$ with r as a cell boundary. Refine it, when needed, at the turning point and inverse branch breakpoints. A partition $\mathcal{D} = \{D_a\}$ of J is* mirror-compatible *if every cell $C_i \subset [0,r]$ has the two cells $C_i^- = -C_i$ and $C_i^+ = C_i$ in $\mathcal{D}$, while cells in $[r,1]$ have only their positive copy. Let P_I and P_J be the row-stochastic exact cell-overlap matrices for T and f respectively:*

$$(P_I)_{ij} = \frac{\mathrm{Leb}(C_i \cap T^{-1}C_j)}{\mathrm{Leb}(C_i)}, \qquad (P_J)_{ab} = \frac{\mathrm{Leb}(D_a \cap f^{-1}D_b)}{\mathrm{Leb}(D_a)}. \tag{5}$$

Let m_0 be the number of paired cells and m_1 the number of unpaired terminal cells. Pullback of a folded observable by q is denoted by J_h; pushforward of a density by q (with the cell-length normalization) is denoted by A_h. These maps are dual under the cellwise Lebesgue pairing.

Theorem 2.2 (Finite folding quotient). *For every mirror-compatible partition,*

$$P_J J_h = J_h P_I, \qquad A_h P_J^{\mathsf{T}} = P_I^{\mathsf{T}} A_h, \qquad A_h J_h = I. \tag{6}$$

Moreover,

$$P_J^{\mathsf{T}} \ker A_h = \{0\}, \qquad \dim\ker A_h = m_0, \tag{7}$$

and

$$\det(zI - P_J) = z^{m_0} \det(zI - P_I). \tag{8}$$

Thus the two matrices have exactly the same nonzero spectrum, algebraic multiplicities, and Jordan blocks at every nonzero eigenvalue. The zero eigenspace is enlarged by the mirror kernel and is not identified further.

Proof. For a cellwise constant g on the folded partition, $g(q(f(x))) = g(T(q(x)))$. Integrating this identity over each mirror pair gives the first relation in (6). The second relation is its adjoint under the cellwise Lebesgue pairing; the factor of two on a paired cell is exactly the pushforward multiplicity and is absorbed in A_h. A density whose two mirror values are opposite pushes forward to zero. Since $f(-x) = f(x)$, the two inverse-branch contributions of such a density cancel in the next transfer step. This proves $P_J^{\mathsf{T}} \ker A_h = 0$. Choose a complement on which A_h is an isomorphism. In the resulting basis the transpose has a zero block of dimension m_0 and the induced quotient block P_I^{T}, up to an inessential triangular off-diagonal block. The determinant factorization (8) follows, and the Jordan assertion follows from the same block form. $\square$

Corollary 2.3 (What the quotient does and does not transfer). *If an aligned folded matrix has an exact eigenvalue -1, every mirror-compatible full matrix has that nonzero eigenvalue as well. The extra mirror degrees of freedom contribute only zero eigenvalues. This statement does not apply to an aligned partition whose cells are not mirror pairs.*

3 The genuine weak bridge outside the unit circle

Let E_h be conditional expectation onto the folded cell partition, with mesh $h \to 0$, and let $\mathcal{P}_T$ be the Perron–Frobenius operator of T on $L^1([0,1])$. It is positive and L^1-contractive.

Proposition 3.1 (L^1 convergence)**.** *For every fixed $g \in L^1([0,1])$,*

$$\widehat{P}_h g := E_h \mathcal{P}_T E_h g \longrightarrow \mathcal{P}_T g \quad \text{in } L^1. \tag{9}$$

If K is compact in $\{z : |z| > 1\}$, then

$$(zI - \widehat{P}_h)^{-1} g \longrightarrow (zI - \mathcal{P}_T)^{-1} g \tag{10}$$

uniformly for $z \in K$, and

$$\sup_{h\,small} \sup_{z \in K} \|(zI - \widehat{P}_h)^{-1}\|_{1\to 1} \leq \sup_{z\in K} \frac{1}{|z|-1}. \tag{11}$$

Proof. Conditional expectations converge strongly to the identity in L^1 and are contractive. Hence

$$\|E_h \mathcal{P}_T E_h g - \mathcal{P}_T g\|_1 \leq \|E_h(g) - g\|_1 + \|(E_h - I)\mathcal{P}_T g\|_1 \to 0.$$

Both $\mathcal{P}_T$ and $\widehat{P}_h$ are Markov contractions. For $|z| > 1$ their resolvents are the norm-convergent Neumann series $z^{-1}\sum_{n\geq 0} z^{-n} P^n$. The contraction bound gives (11); termwise convergence, followed by a uniform geometric tail bound on a compact K, gives (10). □

Remark 3.2 (The unit-circle gap)**.** *The eigenvalue of interest is -1, on the boundary of the domain in Proposition 3.1. The proposition supplies neither a common strong space nor a contour resolvent around -1. It is therefore not a Riesz-projector convergence theorem.*

4 The terminal deterministic spike

On the terminal band $r < y < 1$, the folded map has one admissible inverse branch $x = ((1 - y)/u)^{1/2}$. Consequently

$$(\mathcal{P}_T \mathbf{1})(y) = \frac{1}{2\sqrt{u}}(1-y)^{-1/2}, \qquad r < y < 1. \tag{12}$$

Consider a uniform terminal pair of cells $C_1 = [1-h, 1]$ and $C_2 = [1-2h, 1-h]$. Their cell averages of the right side of (12) are

$$a_1 = \frac{1}{\sqrt{uh}}, \qquad a_2 = \frac{\sqrt{2}-1}{\sqrt{uh}}. \tag{13}$$

Theorem 4.1 (Deterministic BV projection barrier)**.** *The step function $E_h \mathcal{P}_T \mathbf{1}$ has a jump between the two terminal cells of size*

$$|a_1 - a_2| = \frac{2-\sqrt{2}}{\sqrt{uh}}. \tag{14}$$

In particular, for the standard norm $\|v\|_{\mathrm{BV}} = \|v\|_1 + \mathrm{Var}(v)$,

$$\|E_h \mathcal{P}_T\|_{\mathrm{BV}\to\mathrm{BV}} \geq \|E_h \mathcal{P}_T \mathbf{1}\|_{\mathrm{BV}} \geq \frac{2-\sqrt{2}}{\sqrt{uh}}. \tag{15}$$

The coefficient is $0.4714757998\ldots$*.*

Proof. Integrating $(1-y)^{-1/2}/(2\sqrt{u})$ over C_1 and C_2 gives (13). The variation of a step function includes the absolute value of every adjacent-cell jump, which proves (14). Since $E_h\mathbf{1} = \mathbf{1}$, applying the operator to the unit BV function gives (15). □

Remark 4.2 (Scope of the obstruction)**.** *The bound rules out a uniform deterministic entry on this standard BV component. RH-14's tower/spike language records additional singular atoms, and a future mesh-dependent fractional space could in principle absorb the cell profile. No such space, uniform projector estimate, or contour around* -1 *is present in the frozen sources. The theorem is therefore a precise scoped negative, not an impossibility theorem for every conceivable Banach space.*

5 Why the positive-noise bridge cannot be specialized

The strong–weak results in RH-52 and RH-55 use a joint schedule of mesh and positive noise, with a condition of the form $h = o(\sigma^2)$. The estimate (15) is a deterministic $\sigma = 0$ statement. Substituting $\sigma = 0$ into a hypothesis that requires a positive-noise smoothing scale is invalid; it removes precisely the smoothing that controls the terminal spike. Thus the following implications are not licensed:

$$\text{finite aligned } -1 \text{ mode} \not\Rightarrow \text{continuum isolated } -1,$$
$$\text{positive-noise contour} \not\Rightarrow \text{deterministic contour.}$$

For a merely band-aligned partition with one crossing cell, the quotient maps in Theorem 2.2 are not defined. The phase-dependent leakage identity of RH-367 remains a finite local identity, and phase scans do not repair either the missing mirror quotient or the BV bound.

6 Executable audit

The artifact uses exact rational matrices. A folded cyclic test matrix is extended by two mirror rows per paired cell. The four audit sizes have $(m_0, m_1) = (2,1), (3,2), (4,3), (5,4)$; in every row the observable intertwiner and the mirror-kernel image are exactly zero, and the exact characteristic polynomial is the folded polynomial multiplied by the stated power of z. The spike rows check that $h^{1/2}$ times the jump is constant to floating tolerance. These are finite reproduction checks, not spectral limit evidence.

Table 1: Finite audit ledger (generated from `result.json`).

check	count	status
source-lock files	9	pass
mirror quotient dimensions	4	pass
spike scaling rows	4	pass
named Gates A–E	5	false/open

7 Route verdict and Gate ledger

- Route A is `GO`: Theorems 2.2 and 4.1, together with Proposition 3.1, are a new, independently typed quotient/weak-bridge/scoped-negative package.
- Route B is `STOP_SCOPED`: the quotient is restricted to mirror-compatible finite partitions, the L^1 bridge is exterior to the unit circle, and the natural deterministic strong norm has no uniform bound at the terminal spike.

The five project Gates remain false/open. In particular, this paper does not construct a canonical intrinsic dynamical spectral determinant, a scattering completion, a self-adjoint generator, a von-Mangoldt weighted trace, or an equality with the completed-zeta divisor. It does not identify Riemann zeros, construct a Hilbert–Polya operator, or prove the Riemann Hypothesis.

8 Conclusion

The fold supplies an exact algebraic reduction for a strict cofinal subclass of the RH-367 discretizations. It also supplies a genuine exterior weak resolvent statement. The terminal square-root singularity then pays for a sharp deterministic obstruction to the natural strong-space route. The remaining opening is explicit: a new fractional/tower-adapted strong space, or a theorem for non-mirror partitions, would be needed before a contour around -1 can be discussed.

References

[1] Bin Wang. Boundary-aligned cyclic-ulam structure and phase-local leakage, 2026. RH-367 repository release.

[2] Bin Wang. Intrinsic peripheral residue transfer, 2026. RH-52 repository paper.

[3] Bin Wang. A postcritically finite quadratic realization and cyclic ulam data, 2026. Repository source, commit e7d21f646498d77e1c3213d1e4f35dc8466038ff.

[4] Bin Wang. Square-root parity boundary layer, 2026. RH-14 repository paper.

[5] Bin Wang. Strong–weak riesz cutoff transfer, 2026. RH-55 repository paper.